SNx5HVD308xE Low-Power RS-485 Transceivers

1 Features

- Meets or exceeds the requirements of the TIA/EIA-485A standard
- Low quiescent power
  - 0.3-mA Active mode
  - 1-nA Shutdown mode
- 1/8 Unit load up to 256 nodes on a bus
- Bus-pin ESD protection up to 15 kV
- Industry-standard SN75176 footprint
- Fail-safe receiver (bus open, bus shorted, bus idle)
- Glitch-free power-up and power-down bus inputs and outputs

2 Applications

- Energy meter networks
- Motor control
- Power inverters
- Industrial automation
- Building automation networks
- Battery-powered applications
- Telecommunications equipment

3 Description

The SNx5HVD308xE are half-duplex transceivers designed for RS-485 data bus networks. Powered by a 5-V supply, they are fully compliant with TIA/EIA-485A standard. With controlled transition times, these devices are suitable for transmitting data over long twisted-pair cables. The SN65HVD3082E and SN75HVD3082E are optimized for signaling rates up to 200 kbps. The SN65HVD3085E is suitable for data transmission up to 1 Mbps, whereas the SN65HVD3088E is suitable for applications that require signaling rates up to 20 Mbps.

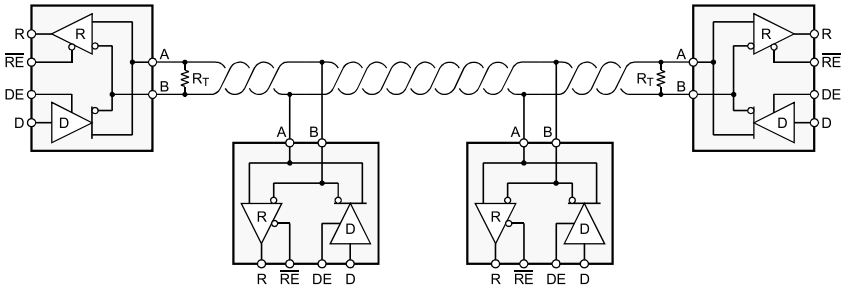
These devices are designed to operate with very low supply current, typically 0.3 mA, exclusive of the load. When in the inactive-shutdown mode, the supply current drops to a few nanoamps, which makes these devices ideal for power-sensitive applications.

The wide common-mode range and high ESD protection levels of these devices, make them suitable for demanding applications such as energy meter networks, electrical inverters, status and command signals across telecom racks, as well as cabled chassis interconnects, and industrial automation networks where noise tolerance is essential. These devices match the industry-standard footprint of the SN75176 device. The power-on-reset circuits keep the outputs in a high-impedance state until the supply voltage has stabilized. A thermal-shutdown function protects the device from damage, due to system fault conditions. The SN75HVD3082E is characterized for operation from 0°C to 70°C and SN65HVD308xE is characterized for operation from –40°C to 85°C air temperature. The D package version of the SN65HVD3082E has been characterized for operation from –40°C to 105°C.

Package Information

PART NUMBER	PACKAGE(1)	PACKAGE SIZE(2)
SN65HVD3082E	SOIC (D, 8)	4.9 mm × 6 mm
	VSSOP (DGK, 8)	3 mm × 4.9 mm
	PDIP (P, 8)	9.81 mm × 9.43 mm
SN65HVD3088E	SOIC (D, 8)	4.9 mm × 6 mm
	VSSOP (DGK, 8)	3 mm × 4.9 mm
	PDIP (P, 8)	9.81 mm × 9.43 mm
	SO (NS, 14)	10.2 mm × 7.8 mm
SN75HVD3082E SN65HVD3085E	SOIC (D, 8)	4.9 mm × 6 mm
	VSSOP (DGK, 8)	3 mm × 4.9 mm

(1) For all available packages, see the orderable addendum at the end of the data sheet.
(2) The package size (length × width) is a nominal value and includes pins, where applicable.



Simplified Schematic

An IMPORTANT NOTICE at the end of this data sheet addresses availability, warranty, changes, use in safety-critical applications, intellectual property matters and other important disclaimers. PRODUCTION DATA.

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4 Revision History

NOTE: Page numbers for previous revisions may differ from page numbers in the current version.

Changes from Revision N (June 2023) to Revision O (July 2023) — **Page**

- Changed pins 11 and 12 to NC in the NS Package Figure 5-2 and Table 5-2 4

Changes from Revision M (February 2022) to Revision N (June 2023) — **Page**

- Changed the Device Information table to the *Package Information* table. Added the NS package....1
- Added the SN65HVD3088E NS package....4
- Added the NS package values to the *Thermal Information, SN65HVD308xE* table....7

Changes from Revision L (November 2021) to Revision M (February 2022) — **Page**

- Added storage temperature T_{stg} to *Absolute Maximum Ratings* table....6
- Changed the *Thermal Information, SN65HVD308xE* table....7

Changes from Revision K (July 2021) to Revision L (November 2021) — **Page**

- Deleted *Feature*: Available in a small MSOP-8 package 1
- Deleted Available in a Small MSOP-8 Package from the title....1
- Changed the ψ_{JT} D package value from 78.8 to 8.8 in the *Thermal Information, SNx5HVD3082E* 7

Changes from Revision J (October 2017) to Revision K (July 2021) — **Page**

- Changed the *Thermal Information* tables....7

Changes from Revision I (September 2016) to Revision J (October 2017) — **Page**

- Changed 3.3 V to 5 V on the V_{CC} pin in Figure 9-4 21

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